Final Project Proposal

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Submission Date: 11/05/2026

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1. Title

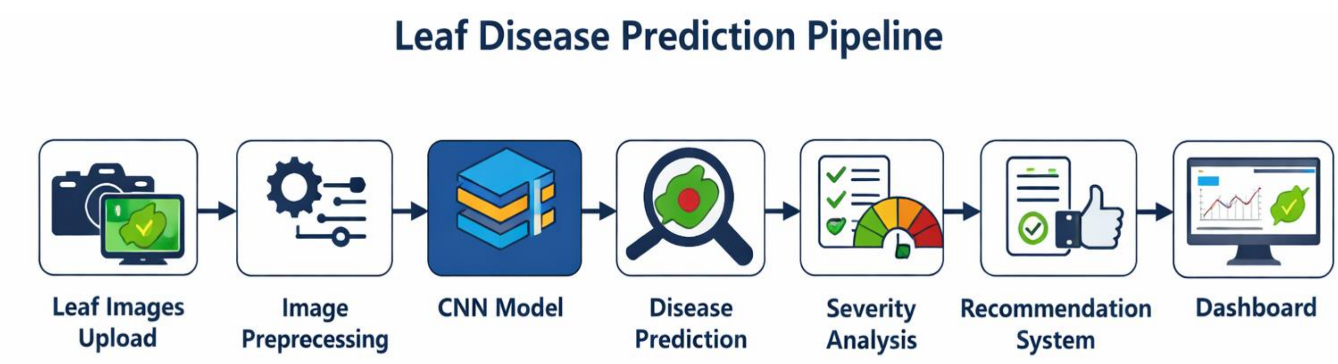
AI-Powered Coconut Leaf Disease Detection and Health Monitoring System

Project Overview

The proposed project focuses on developing an AI-powered web-based system capable of detecting coconut leaf diseases using deep learning and computer vision technologies. Coconut leaf diseases can significantly reduce plantation productivity and economic value if they are not identified during the early stages. Traditional disease detection methods mainly depend on manual inspection, which is often time-consuming and inaccurate.

The proposed system allows users to upload coconut leaf images and receive real-time disease predictions, severity analysis, and treatment recommendations. The system uses Convolutional Neural Networks (CNNs) with transfer learning models such as MobileNetV2 to improve classification accuracy and performance.

In addition, the system includes a plantation health monitoring dashboard for analyzing disease statistics, prediction history, and plantation conditions. The system aims to support farmers and agricultural officers by providing a fast, affordable, and user-friendly disease monitoring solution.



2. Introduction

Agriculture plays an important role in the economy of many tropical countries, and coconut cultivation is one of the major agricultural industries in Sri Lanka. Coconut trees provide several economic benefits through products such as coconut oil, coconut milk, and fiber. However, coconut plantations are highly affected by leaf diseases that can reduce crop quality and productivity.

Traditional disease detection methods mainly rely on manual inspection by farmers or agricultural experts. These methods are often slow, inaccurate, and dependent on human experience. In many cases, diseases are identified only after severe symptoms appear, causing significant plantation losses.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies have enabled the development of smart agricultural systems. Convolutional Neural Networks (CNNs) can automatically identify disease-related patterns such as discoloration, texture changes, and infected regions from leaf images with high accuracy.

This project proposes an AI-Powered Coconut Leaf Disease Detection and Plantation Health Monitoring System that uses deep learning techniques for accurate disease classification and plantation monitoring. The system also provides disease severity estimation, treatment recommendations, and prediction history through a web-based dashboard developed using React, Tailwind CSS, and Flask.

3. Current situation leading to problem identification

3.1 Current Situation

Currently, coconut leaf diseases are mainly identified through manual inspection by farmers or agricultural officers. Disease detection is based on visible symptoms such as yellowing leaves, dark spots, and drying patterns. This method is time-consuming, inaccurate, and highly dependent on human experience.

In large plantations, regular monitoring becomes difficult, causing diseases to spread before proper treatment is provided. Small-scale farmers also face difficulties in accessing agricultural experts for accurate diagnosis.

3.2 Problems in Existing Methods

Traditional disease detection methods have several limitations:

- Delayed disease identification.
- Human errors during inspection.
- Difficulty in monitoring large plantations.
- Increased crop and financial losses.
- Lack of automated monitoring systems.

Although some AI-based systems exist, many focus only on image classification and do not provide plantation monitoring or user-friendly web access.

3.3 Problem Statement

Currently, there is no affordable AI-powered web-based system capable of accurately detecting coconut leaf diseases while supporting plantation health monitoring and disease management.

3.4 Proposed Direction Toward a Solution

The proposed system uses CNN-based deep learning techniques to detect coconut leaf diseases from uploaded images. The system also provides severity analysis, treatment recommendations, and plantation monitoring through a web-based dashboard.

Comparison of Existing Methods and Proposed Solution

Existing Methods	Limitations
Manual Inspection	Human error and slow detection
Traditional Monitoring	Difficult to monitor large plantations
Basic AI Systems	Limited features and accessibility
Proposed System	AI detection with monitoring dashboard

4. Proposed technique to solve the current problem

4.1 Overview

The proposed system is an AI-powered web application designed to detect coconut leaf diseases using deep learning and computer vision techniques. The system allows users to upload coconut leaf images and receive real-time disease predictions along with severity analysis and treatment recommendations.

The project uses Convolutional Neural Networks (CNNs) with transfer learning models such as MobileNetV2 to improve prediction accuracy and reduce training time. The system is designed to provide a fast, affordable, and user-friendly solution for farmers and agricultural officers.

4.2 Technologies Used

- Frontend: React.js and Tailwind CSS
- Backend: Flask
- AI Model: TensorFlow / Keras
- Image Processing: OpenCV
- Database: Firebase or MySQL

4.3 System Workflow

The system follows several main stages:

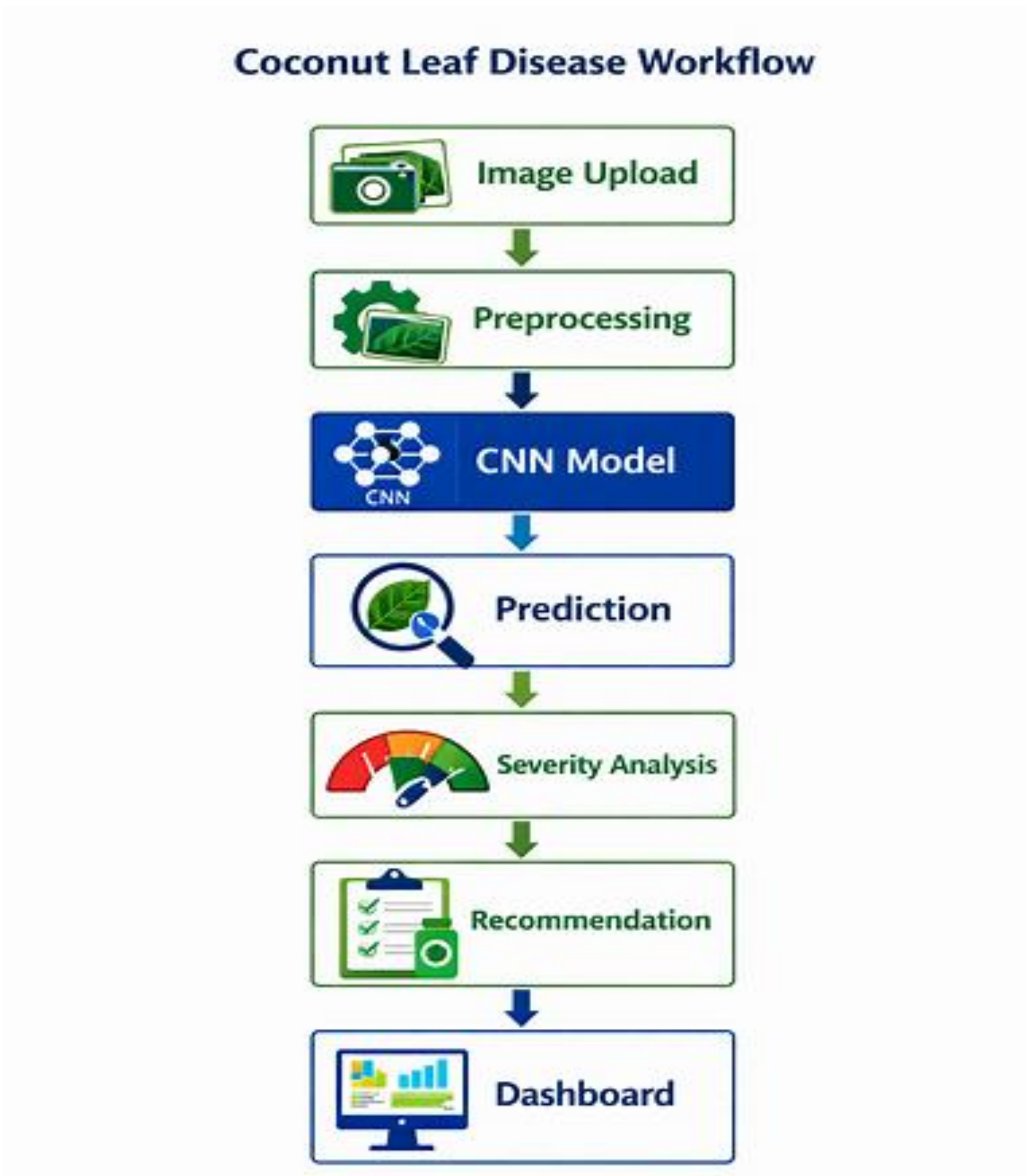
1. Image Upload – Users upload coconut leaf images through the web application.
2. Image Preprocessing – Images are resized and normalized before prediction.
3. Disease Detection – The CNN model analyzes the image and predicts the disease category.
4. Severity Analysis – The system estimates the infection level as Mild, Moderate, or Severe.
5. Recommendation System – Treatment and prevention suggestions are provided.
6. Dashboard Monitoring – Prediction history and disease statistics are stored and displayed.

4.3 Expected Features

- Real-time disease prediction.
- Image-based disease analysis.
- Severity estimation.
- Plantation health monitoring dashboard.
- Treatment recommendations.
- Prediction history tracking.

Main System Components

Component	**Purpose**
React + Tailwind	User interface
Flask	Backend API
CNN Model	Disease prediction
OpenCV	Image preprocessing
Dashboard	Monitoring and analytics



5. Feasibility analysis

The feasibility of the proposed Coconut Leaf Disease Detection and Plantation Health Monitoring System can be evaluated through technical, economic, and operational feasibility. These factors help determine whether the project can be successfully developed and implemented using the available resources.

A. Technical Feasibility

The project is technically feasible because it uses widely available technologies such as TensorFlow, OpenCV, Flask, React, and Tailwind CSS. The Kaggle Coconut Leaves Disease Dataset provides sufficient image data for training and testing the CNN model.

The system can also be developed using standard hardware with cloud GPU support through platforms such as Google Colab or Kaggle Notebooks.

B. Economic Feasibility

The project is cost-effective because most tools and technologies used are open-source and freely available. Development platforms such as Google Colab, VS Code, and GitHub reduce implementation costs.

The proposed system may also help reduce plantation losses by enabling early disease detection and better disease management.

C. Operational Feasibility

The system is designed to be simple and user-friendly for farmers and agricultural officers. Users only need to upload coconut leaf images to receive predictions and recommendations.

The web-based dashboard allows users to monitor plantation conditions and prediction history easily through a standard web browser.

Feasibility Analysis Summary

Feasibility Type	**Description**
Technical Feasibility	AI and web technologies are available and easy to implement
Economic Feasibility	Uses low-cost and open-source tools
Operational Feasibility	Simple and user-friendly web system

6. Project description

6.1 Overview

The proposed system is a web-based AI application developed to detect coconut leaf diseases and support plantation health monitoring. The system uses deep learning and computer vision techniques to analyze coconut leaf images and provide accurate disease predictions.

Users can upload coconut leaf images through the web application and receive real-time predictions, severity analysis, and treatment recommendations. The system is designed to help farmers and agricultural officers identify diseases quickly and reduce plantation losses.

6.2 Main Functional Modules

6.2.1. Image Upload Module

This module allows users to upload coconut leaf images through the web application. The uploaded images are validated before being processed by the AI model. The interface is designed to be simple and user-friendly so that farmers and agricultural officers can easily use the system without technical knowledge.
Main functions:

- Upload leaf images
- Validate image format
- Preview uploaded images
- Send images for processing

6.2.2. Disease Detection Module

The Disease Detection Module is the core component of the system. It uses a Convolutional Neural Network (CNN) model with transfer learning techniques to identify coconut leaf diseases from uploaded images.

The model analyzes features such as:

- Color changes
- Dark spots
- Leaf discoloration
- Texture abnormalities
- Infection patterns

After analysis, the system predicts the disease category and displays the confidence score.

Example:
Leaf Spot Disease – 92% Confidence

6.6.3. Severity Analysis Module

This module estimates the severity level of the detected disease based on the infected area visible on the leaf image.

Severity levels include:

- Mild
- Moderate
- Severe

The severity analysis helps users understand the seriousness of the infection and supports better treatment planning.

6.6.4. Recommendation Module

Based on the prediction result, the system provides treatment recommendations and prevention methods to help users manage diseases effectively.

Recommendations may include:

- Fertilizer suggestions
- Fungicide recommendations
- Leaf removal guidance
- Disease prevention methods
- Plantation maintenance advice

This module improves the practical usefulness of the system for real-world agricultural applications.

6.6.5. Dashboard Module

The dashboard module provides plantation monitoring and analytical features. All prediction records are stored and displayed through visual charts and statistics.

Dashboard features include:

- Prediction history
- Disease frequency analysis
- Severity distribution
- Plantation monitoring charts
- Confidence score tracking

The dashboard helps users monitor plantation conditions and identify common disease patterns over time.

7. Deliverables

The project deliverables represent the final outputs and components produced during the development of the Coconut Leaf Disease Detection and Plantation Health Monitoring System.

Main Deliverables

1. Trained CNN Model

A deep learning model capable of accurately detecting coconut leaf diseases from uploaded images.

2. Web-Based Application

A responsive web application developed using React, Tailwind CSS, and Flask for disease prediction and plantation monitoring.

3. Disease Severity Analysis Module

A module capable of estimating disease severity levels such as Mild, Moderate, and Severe.

4. Recommendation System

A feature that provides treatment methods and disease prevention recommendations based on prediction results.

5. Plantation Monitoring Dashboard

An analytical dashboard displaying disease statistics, prediction history, and plantation monitoring information.

6. Preprocessed Dataset

A cleaned and organized coconut leaf image dataset prepared for CNN model training and evaluation.

7. Project Documentation

Complete project documentation including proposal, system design, testing results, and final report.

8. Resources required

The development of the proposed system requires several hardware, software, and development resources to support model training, web application development, and system deployment.

Hardware Requirements

The project can be developed using a standard laptop or desktop computer with moderate specifications.

Recommended hardware includes:

- Intel i5/Ryzen 5 processor or higher
- Minimum 8GB RAM
- SSD storage
- Smartphone or camera for capturing leaf images

Software Requirements

Several software tools and libraries will be used during development.

Development Technologies

- React.js
- Tailwind CSS
- Flask
- Python

AI and Image Processing Libraries

- TensorFlow / Keras
- OpenCV
- NumPy
- Pandas
- Matplotlib

Development Platforms

- VS Code
- GitHub
- Google Colab
- Kaggle Notebooks

Dataset Resource

The project uses the [Kaggle Coconut Diseases Dataset](https://www.kaggle.com/datasets/samitha96/coconutdiseases/data), which contains multiple coconut leaf disease images collected under different environmental conditions.
https://www.kaggle.com/datasets/samitha96/coconutdiseases/data

9. Expected output and outcome

Expected Output

The proposed system is expected to generate several useful outputs related to coconut leaf disease detection and plantation monitoring.

1. Disease Prediction

The system will classify uploaded coconut leaf images into disease categories such as healthy leaves and infected leaves.

2. Confidence Score

Each prediction will include a confidence percentage indicating the accuracy of the prediction.

3. Severity Analysis

The system will estimate the disease severity level as Mild, Moderate, or Severe.

4. Recommendation System

Treatment suggestions and disease prevention methods will be provided based on the detected disease.

5. Monitoring Dashboard

The dashboard will display prediction history, disease statistics, and plantation monitoring analytics.

Expected Outcome

The proposed system is expected to improve early disease detection and support better plantation management. Farmers and agricultural officers will be able to identify diseases faster and take preventive actions before diseases spread across plantations.

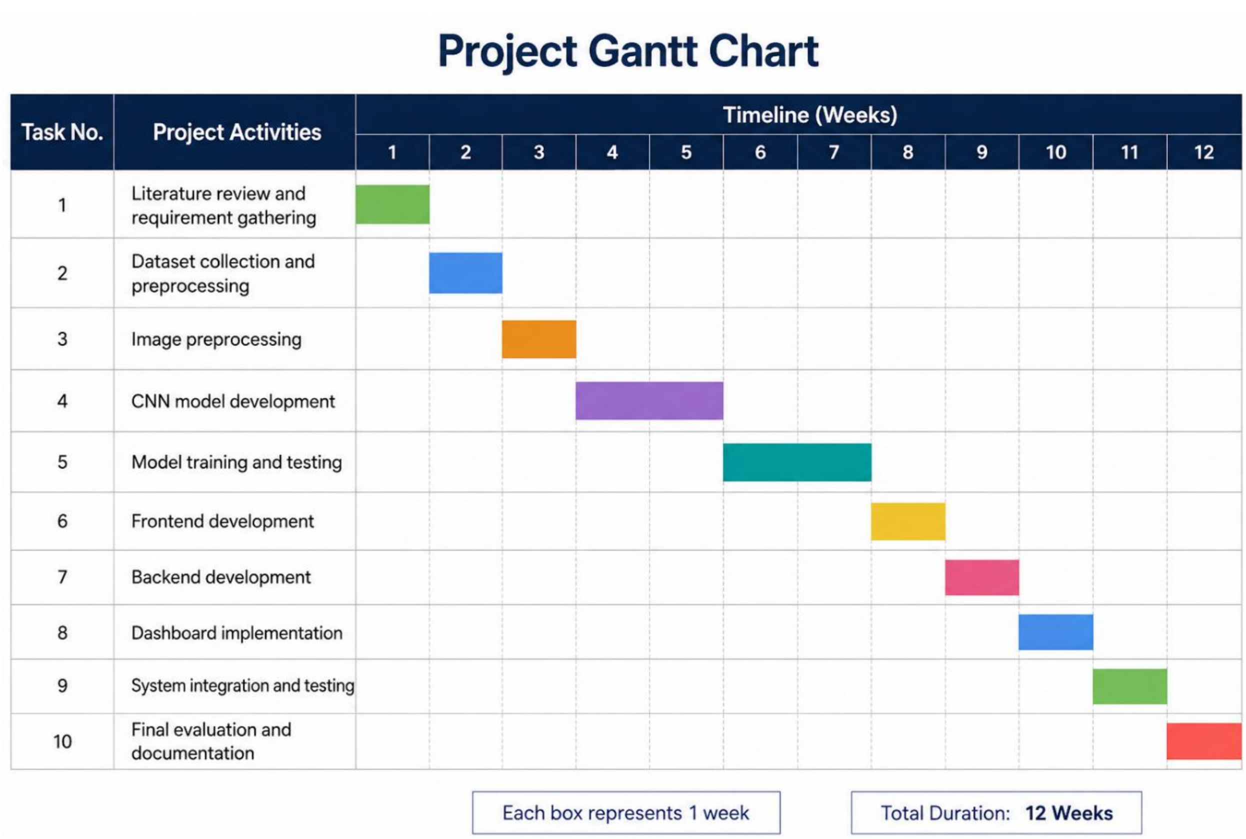
The project also demonstrates the practical use of Artificial Intelligence and Deep Learning technologies in agriculture while reducing plantation losses and improving productivity.

10. Time plan for implementation

The project will be completed over a 12-week period, divided into structured phases covering data collection, model development, testing, and reporting. This plan ensures organized progress and proper time management.

Project Activities

- Literature review and requirement gathering
- Dataset collection and preprocessing
- CNN model development and training
- Frontend and backend development
- System integration and testing
- Dashboard implementation
- Final evaluation and documentation



11. Limitations

Although the proposed Coconut Leaf Disease Detection and Plantation Health Monitoring System provides several advantages, certain limitations may affect system performance and implementation.

1. Limited Disease Categories

The current dataset contains a limited number of coconut leaf disease categories. Therefore, the system may not accurately identify diseases that are not included in the training dataset. Future improvements can include expanding the dataset with additional disease classes and real-world samples.

2. Image Quality Dependency

The prediction accuracy may be affected by poor image quality, including blurry images, low lighting conditions, or complex backgrounds. Proper image preprocessing and dataset augmentation techniques will be used to reduce this limitation.

3. Environmental Variations

Different environmental conditions such as lighting, shadows, and camera angles may influence prediction performance in real-world scenarios. Training the model using diverse images can improve model robustness.

4. Hardware Constraints

Training deep learning models may require higher computational resources and GPU support. Cloud-based platforms such as Google Colab and Kaggle will be used to overcome hardware limitations.

5. Model Generalization

The trained model may perform differently on unseen real-world data compared to the training dataset. Continuous testing and dataset improvements can help improve model generalization and prediction accuracy.

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